

Chapter 3

Sharing IP Strategy: Commercialization

The previous chapter explored how IP legislation affords patent and copyright holders with a legal monopoly over their intellectual asset; with a view to reducing the amount of uncertainty they have over them. Other IPRs also are intended to grant exclusive control over valuable features of those assets. However, the market power allowed by such legal title is far from giving absolute certainty and control to title-holders. Market conditions limit their real capacity to impose full control to their asset use, due to the qualifying circumstances the market presents to them. In principle, that situation leads IP holders to deploy strategies aimed at enhancing their power over the market, to impose economic limitations upon their rivals on their capacity to seize strategic competitive capabilities.

Alternatively, companies may prefer to trade strategic assets in order to develop their own competitive capabilities. This option allows them to obtain useful inputs that may allow them to develop competitive advantages needed to penetrate new markets. IP managers define what skills they may develop in cooperation with other businesses; and how their patent titles give them intangible assets they can trade with other firms as currency to obtain missing capabilities, either by trading them or by licensing them in exchange for other licenses (patent pooling). This chapter will explore in detail how these trades take place.

3.1 Drawbacks of Imposing IP's Market Power

Patent selling and licensing are the most popular instruments for commercializing IP. The IP licensing market has grown dramatically in the last 20 years worldwide. According to WIPO, from 2000 to 2015, IP licensing grew from 100 billion to approximately 500 billion. Patent licensing is by far the most popular IP traded, but also trademarks and copyrights enjoy an important share of the market. This situation is not surprising. IP licensing is perceived as an expedient way for IP holders to develop a competitive advantage, by cooperating with competitors, trading their

knowledge, as opposed to imposing their market power. Yet, IP licensing or sale has a tactical advantage that makes it very appealing, as we will see in this chapter.

Innovators may find it convenient to leave the hazards and uncertainties of commercializing IP in the hands of patent brokers and focus on the innovation activity. The way to do this is by selling or licensing the patent to a firm in need of such knowledge to develop a productive capability or to trade their IP portfolio (other than patents), either through a patent broker or directly to the buying or licensee company.

Alternatively, the innovator may prefer to retain a relationship with the business party, in the interest of fulfilling some other strategic corporate goal. For example, the innovator may be a researcher interested in using the business' facilities or teams to develop future research or may find it useful to obtain financial resources to carry out alternative scientific research, or may be interested in getting access to the network of contacts and financial resources available to the business party. An innovator may discover that he either needs to license technology from his company to augment his product development efforts or needs to create a new revenue stream by licensing technology that he has developed for other people. In cases such as these, a licensing agreement would enable the mechanism to do this, through a contractual arrangement between the innovator (licensor) and the business partner (licensee) in exchange for an agreed payment (fee or royalty).

According to Fisher and Oberholzer-Gee (2013), three factors may increase the costs of imposing market power for IP owners, thereby driving them to trade with potential competitors, instead. These are reasons why they may find it more convenient to share their IP with rival companies either through licensing, selling or even giving away their assets.

First, IP owners obtain market power from their legal entitlements, but their capacity to extract "rent" greatly depends on the strategic behavior of competitors. Competitors' actions may diminish the overall value of the market where IP owners expect to obtain their IP revenues. Faced with this contingency, IP owners may find it convenient to negotiate an understanding with their competitors to share the income resulting from their IPRs. The cutthroat competition between Procter and Gamble Company (P&G) and Colgate-Palmolive Company in the home teeth-whitening industry highlights this point. P&G introduced Crest Whitestrips in 2000, offering consumers a less expensive method to whiten their teeth. This approach had patent protection that made it very hard for Colgate to invent around. To Colgate's dismay, Crest began to rapidly gain market share, not only on the whitening strips but also across its portfolio of dental products traded under the Crest brand. To halt Crest's growth, Colgate introduced a less useful product at a lower price, thereby triggering a price war that dissipated much of the gains the market offered to both P&G and Colgate. Arguably, Crest could have retained a much larger chunk of the profits by adopting less aggressive marketing tactics and developing an entente with Colgate.

A second factor that justifies a sharing strategy is to prevent competitors from "inventing around." Again, here is a case where the exercise of market power through IP undermines the long-term profitability of the right's owner. This is particularly visible where the owner would want to transform its technology as the dominant "standard" of the industry at the early stage of the sector's development. One of the

most well-known cases in point is Apple's self-defeating commercialization strategy in the 1980s, and the lessons drawn for their strategy in the 2000s over the iPhone.

In the early 1980s, Apple and Microsoft were partners in the development of the Macintosh. Apple required innovative software to develop its upcoming operating system iOS; Steve Jobs was to focus Apple on the computer business, thus leaving software development to Microsoft. By owning the patents that helped her to control the OS, Apple expected to impose her market power upon consumers and make customers pay accordingly, as well as managing software developers like Microsoft. However, Microsoft had other thoughts.

Apparently, Bill Gates did not want Microsoft to follow Apple's plan, as this would have made it dependent from her ordering. It realized that the Graphical User Interface was the future of computing, so he did not want Microsoft to be a small software developing company. Devising a strategy to avoid technological oblivion, Microsoft put their efforts into developing its own Graphical OS (Windows), capable of competing with Apple in the upstream market. However, to do that, they had to make this OS as widely available as possible. They allied with IBM, who was desperate to break into the personal computing market; so Microsoft reached an agreement with IBM to outsource their Windows OS on IBMs computers. Since IBM had wider brand recognition and reach among distributors, Microsoft immediately gained a competitive advantage over Apple. Also, Microsoft started selling its OS to any PC maker that would buy it. By contrast, Apple was foreclosing her possibilities, by imposing a market power strategy upon her distributors, who thereby increasingly started switching to IBM/Windows, and subjecting consumers to pricey offerings and technological idleness (it took Apple 7 years to produce Apple III, the sequel to Apple II). Customers and developers were familiar with the IBM brand and bought IBM computers. As Microsoft grabbed market share, developers started working on applications for its OS instead of Apple's Macintosh system. Microsoft became the dominant firm from then on. At the end of the 80s, Apple was primarily a minor player in the computer industry. The lesson is that they should have licensed in more favorable terms, until becoming the industry standard (which they never were, contrary to the general opinion).

Many people thought history would repeat itself with Apple's iPhone in the wake of Google's cheaper Android arrival. However, this time things were different for Apple. Apple's IP strategy was much more robust, as it had spent years building brand recognition through the iPod and the Mac and its constant innovation and development of product portfolio with the iPad.

As Yarow (2012) explains:

Each year Apple has released a phone that is better than the phone it sold the year before. There was no seven-year opportunity for a rival to release a phone that was better than the iPhone.

Also, unlike its previous experience, Apple had no real challenger, comparable to IBM who could quickly switch into the personal computer market by just finding the right OS provider. When Google got into the smartphone industry, it was starting from scratch with Android. In 2007 Google was still widely unknown to most

people. It was not a trusted brand like IBM. In short, this time Apple had to learn how to deploy a successful IP management strategy that could sustain its costlier licensing program, and that prevented competitors from inventing around.

The third factor is the development of network effects that lead to smaller markets for complementary goods. A product or services value increases with the number of users, as it happens with social networks: the value of the standard or network increases as more individuals use it (“direct network effects”). However, the value of the platform also increases as the number of users grows because a greater number of complementary products and services become available. Complements are products that increase the demand for another product.

For example, some readers may remember back in 2007 when Apple first told developers that to develop for the iPhone, they would need to build WebApps for Safari. Walter Isaacson (2011) recalls how Steve Jobs stubbornly opposed the idea of third party apps and wanted to create web apps that iPhone clients could only use through the iPhone’s mobile Safari web browser. Art Levinson, an Apple board member, recalls that Jobs was against third party apps partly because he felt they could not handle all the complexities that would be involved in policing third-party app developers. This is a usual case of a top-down approach towards innovation that prioritizes policing the process as opposed to leaving it unleashed.

So, here we have the man reckoned to be the personification of innovation in the personal computer industry, the man regarded as *the* guru of business trends in the digital industry, who failed to see the trend of the market. This trend was driving the market towards making the iPhone a carrier for the development of apps, whether WebApps for Safari or not. He lacked a proper IP management strategy to capture profits arising from his IP over the iPhone’s iOS mobile operating system, in a way that could not only guarantee Apple’s market power exploitation as a mobile telecom carrier but also as an operating platform for the development of thousands of apps.

3.2 Contractual Flexibility in Licensing and Selling Patents

Depending on the specifics of the transaction desired, a variety of licensing agreements are available. These include Technology License Agreements, Trademark Licensing and Franchising Agreements, and Copyright License Agreements. In practice, these agreements may involve different types of IP, so it is usual for the parties to negotiate them concurrently.

Licensing is very flexible to adapt to whatever degree of business integration both the licensor and licensee desire. Usually, provisions enabling adaptation of capabilities between both parties include the following:

- (a) *Subject matter*. What is the asset that the parties agree to license? Is it a patent, copyright, an image, a computer program, a song or a recipe?
- (b) *Rights to be granted or conveyed*. License agreements allow the right to use, to sell, to make, or to have made by a third party the licensed IP. The parties will want to be certain to get the rights they need.

- (c) *Exclusivity*. Is the license exclusive, i.e. is the licensee the only one authorized to use the intellectual asset? Is it non-exclusive or is it partially exclusive? The licensee may find it valuable to hold an exclusive license in markets where there are no competing technologies; also exclusivity may be needed if the technology still requires considerable development from the licensee.
- (d) *Geographical scope*. Are there any geographic limitations such as regional, national or local?
- (e) *Field-of-use restrictions*. A licensor may find it relevant to license the technology for use in one industry but not another. A nanotechnology developed for rapidly detecting changes in the composition of proteins may be used selectively in several markets where the technology may be applied, such as vector diagnostics and water or food contamination. Thus, it may be licensed to different licensees depending on their particular capabilities and roles in complementing the value offered by the licensed intellectual asset.
- (f) *Sublicense*. Does the licensee have rights to license the technology again? These rights may be necessary if the licensee plans to incorporate it into her products.
- (g) *Assignment rights*. If the licensee company gets bought, does the license get canceled or does it get assigned to her acquirer?
- (h) *Payment*. Payment could adopt the form of an upfront fee, or through running royalties based, typically, on the sales of the product. One could negotiate a combination of these payment modalities. Choosing what payment system is tricky if the intellectual asset to be licensed gets embedded into a larger product that is sold by the licensee. How are royalties calculated then? What if the licensee does not charge for her product? Usually, startups will pay the upfront fee with equity, because they typically don't have much cash.
- (i) *Diligence*. The licensor may impose on the licensee company duties to commercialize the technology within a certain timeframe or by showing some milestones.
- (j) *Patent litigation*. Sometimes licensors require licensees to protect the technology by paying litigation expenses in case of infringement by third parties.
- (k) *Confidentiality*. Clauses requiring confidentiality are standard to protect know-how implicit in the intellectual asset.
- (l) *Improvements*. Does the licensor yield to the licensee rights on any improvements developed by the latter?

The flexibility allowed by the contractual mechanisms offered by the licensing and sale of intellectual property is the most obvious way of dealing with cognitive problems arising from the IP system. The parties' cognitive expectations about potential business outcomes resulting from their relationship drive their actions towards specific business goals.

As WIPO (2015) indicates:

All of these mechanisms either on their own or in combination will provide a startup, as a licensor or licensee, a wide variety of possibilities in conducting business in your own country or elsewhere. As an intellectual property owner and a licensor, your startup can

expand its business to the frontiers of your partners' business and ensure a steady stream of additional income. As a licensee, your startup can manufacture, sell, import, export, distribute and market various goods or services which it may be prevented from doing otherwise.

The complexity of the IP system creates new challenges that cannot be solved only by contractually resorting to patent licensing or selling. Indeed, IP commercialization strategies do not need to involve patent registration. In fact, of their total investment on innovation, firms seldom appeal to patent registration due to its high costs and convoluted process. Disclosure requirements put innovators in a difficult position: trusting the effectiveness of an expensive enforcement mechanism to redress any infringements against their entitlement by third parties.

More often, firms resort to a combination of strategies, and a portfolio of IP protection that extends well beyond the scope of patent registration.

The threat of patent litigation arising from the lack of clarity in the assignment of patent claims increases with the level of complexity featuring the industry concerned. That could discourage businesses from making their inventions public through the patent registry, and they therefore may prefer to rely on commercial or industrial secrets or to develop preventive strategies such as the reciprocal licensing of patents to avoid traffic jams ("patent thickets"), which may restrict competition.

3.3 Forms of Patent Trading: Brokerage and Pooling

Trading parties in the IP market have developed a myriad of ways in their attempt to furnish such transactions with information needed for the exchanges to take place. We will examine in this section only two primary forms of IP trading: patent brokerage and patent pools. For simplicity's sake, and given their higher liquidity in the market, we will assimilate patent trading to IP trading, which covers a broad range of IPRs different from patents, such as copyrights, trade secrets, trademarks and utility models.¹

First, patent brokerage plays an important role in reducing transaction costs between producers and consumers of IP by supplying specialized information to the market in the form of specialized valuation services. Given the difficulties of attaining a patent market value, broker's experience is critical to determine the most accurate value achievable under these exceptional circumstances. They also are helpful to bring useful third party information (information about the industry, the technology involved, the competitors, and so on) to the negotiation table. This aid could be critical to reducing information asymmetries between a seasoned firm on the one hand, and a smaller first-time seller, on the other.

In the past, brokers were lawyers and patent agents who matched inventors looking to sell new technologies to investors or buyers seeking to commercialize them. However, the booming development of the patent registration has made this work

¹ Millien and Laurie (2009) find 17 highly particular types of intermediaries, including distinctions between Patent Licensing and Enforcement Companies, and IP/Technology Development Companies.

more difficult. Today, technologies overlap and are valuable depending on the size of the portfolio where they belong. Wang (2010: 164) acknowledges: “the new focus on patent assets has not corresponded to a rise in sophistication among parties to intellectual property transactions.”

Perhaps this explains why patent brokers’ presence in the market is limited; they are small in scale and tend to focus on facilitating high-end licensing transactions bearing hefty price tags (10% of the value of the deal and sometimes 20–30%).

Although some authors (Wang 2010) seem to suggest the recent emergence of the market for IP, and hence the novelty of these patent intermediaries, the truth is that brokers have existed since the very inception of the patent system. Lamoreaux and Sokoloff (2003) account for the positive effect of agents on the US market for patented technology between 1870 and 1920. Today it includes players such as Inflexion Point Strategy,² iPotential (Epicenter Group),³ Ocean Tomo,⁴ IP Value⁵, Pluritas⁶ and Thinkfire⁷ (Wang 2010: 171; Hagiu and Yoffie 2013).

The second form of patent commercialization, patent pooling, may be best grasped by looking at the experience of Oracle Corporation, one of the most innovative firms in the software industry. This company refused to patent its inventions to avoid the costs of potentially infringing a myriad of existing competing patents related to the technologies Oracle applies while producing complex software products. In fact, Oracle did not obtain its first patent until 1995 and acquired only 161 patents during the 1990s, although patents have been issued on database techniques at least since the early 1970s, and especially since 1990.

In the course of the USPTO Hearings in 1994, Jerry Baker, Senior Vice President of Oracle Corporation, the software firm specialized in innovative database management systems explained why, despite all their innovative capabilities, he had consciously decided not to give priority to patent applications to protect their technologies.

Of course, not all technology firms have Oracle’s advantage of an established brand name in the market, so they have to rely on patents, rather than other intangible rights. Their options are limited to finding some way of avoiding the potential clash of patent claims whose technology boundaries may collide or overlap. In the wake of patenting proliferation since the 1980s, this is a very real problem. “Patent thickets” resulting from this phenomenon have created “a dense web of overlapping intellectual property rights that a company must hack its way through to commercialize new technology”⁸ (Shapiro 2001). Alternatively, in other words, “an overlap-

²<http://www.ip-strategy.com/>

³<http://www.IPotential.com>

⁴<http://www.oceantomo.com/>

⁵<http://www.ipvalue.com/>

⁶<http://www.pluritas.com/>

⁷<http://www.thinkfire.com/>

⁸Patent thickets may have ambiguous effects on patent transactions. On the one hand, they may discourage patent deals because patent ownership is not centralized but scattered among many patent holders. As the number of patent owners with whom buyers and licensees have to strike deals increases, transaction costs rise. However, they may also encourage licensing deals because

ping set of patent rights.”⁹ In the absence of patent pools, patent thickets require firms seeking to commercialize new technologies to negotiate individual patent licenses with multiple patentees. That gives rise to as many conflicting claims of intellectual property ownership as there are patent owners.

Thus, patent thickets create a problem of royalty stacking. In this situation multiple parties that hold market power in a chain of production are collectively forced to increase their prices above the level they would have in the absence of the other firms with overlapping technologies. In other words, stacking reduces their joint profits (Hagiu and Yoffie 2013: 50). Harper (2015) notes that patent thickets reveal an “anticommons problem” that induces innovative firms to underuse the patented invention because they are faced with their competitors’ veto since no firm has an existing right to use the IP. In his view, this situation arises because IP owners apportion IPRs into too many small fragments, so it creates prohibitive asymmetric transaction costs that make it harder to regroup them into alternative bundles, according to the needs of entrepreneurs and end-users.

The solution is agreeing to cross-license each other’s technology; hence the rationale of “patent pools.” A patent pool is an agreement between two or more companies to cross-license their patents relating to a particular technology. The consortium that runs patent pools sets forth the rules whereby participants submit their IP to a joint venture. This joint venture helps firms developing highly sophisticated technologies protected with different layers of patent claims to share and transfer intellectual property and, ultimately, facilitate innovation with a minimum of disturbance from IP-related legal issues.

Sometimes SMEs create unique forms of patent pooling around standard-setting institutions to raise their chances of finding third party clients interested in their technology. Standard-setting organizations endorse a particular technological standard: to execute this arrangement, SMEs members agree to cross-license or to pay royalties to the single standard owner (Hagiu and Yoffie 2013). In other words, several pooled patents set down a standard that increases the potential value of individual patents by raising probabilities of trading them under the norm umbrella. By reducing transaction (negotiation) costs between small companies, the standard enables more transactions to occur between the pool and potentially interested third party clients (licensees). This way of conducting patent pooling has limitations, however, as the number of industries and technologies involved is relatively small. Therefore, their significance in terms of total patent market share is relatively lower.

the presence of overlapping patent rights may reduce the value at stake in each patent licensing negotiation. Furthermore, incumbent firms (especially in mature industries) may use patent thickets as a strategy to raise the entry costs of potential competitors, if patent standards are low, and the technologies are complex (Bessen 2002). Patent thickets are thus used to defend against competitors designing around a single patent. When they assert these patents through the threat of litigation, innovators are forced to share rents under cross-licenses, making R&D incentives sub-optimal. On the other hand, when lead-time advantages are significant, and patent standards are high, firms pursue strategies of “mutual non-aggression.”

⁹The expression “thicket” comes from the *SCM Corp. v. Xerox Corp.* patent litigation case (*SCM Corp. v. Xerox Corp.*, 645 F.2d 1195, 1203 (2d Cir. 1981), wherein SCM argued that that Xerox assembled a “patent thicket” to prevent competition.

Closson (2009) explains the historical development of the patent pools, associated with the technological development of competing technologies used in sewing machines. In 1856, four companies¹⁰ held the patents related to the sewing machine; however, they employed much of their endeavors suing each other for patent infringement. Accordingly, they resolved to enter their patent portfolios into a technology pool (held in a trust) that would allow each company to freely cross-license the technology from the pool. In this way, they saved resources from litigation to use instead for manufacturing and marketing sewing machines.

Patent pools are difficult to form because they require the participation of all key patentees. If they are not in, it is likely that the pool will not be sustainable. For this reason, sometimes the government has stepped in and forced the creation of a pool, to preserve resources from being squandered in wasteful lawsuits. As an example, in 1917 the US Secretaries of War and Navy ‘recommended’ (i.e., forced) the Wright Company and the Curtiss Company, two principal owners of essential airplane patents, to set up a patent pool. At the time, the United States was about to join the First World War. Until the brink of the war, multiple patent suits had prevented new airplane construction. On this ‘recommendation’ coming from the US Federal Government, Wright, Curtiss and other essential patentees formed the Manufacturers’ Aircraft Association. Of course, in the arms industry the government usually has a high leverage to stir things following the dictates of “national security.”

Other historical examples of patent pools include several industries. For example, in the film industry, in 1908, Armat, Biograph, Edison and Vitagraph agreed to assign “all the patents in the early-day motion picture industry.” The agreement also stated how much royalties movie exhibitors would pay to the pool as licensees (Wanamaker 2003; Aberdeen 2005). In 1916, in the consumer goods industry, the owners of various folding beds patents agreed to provide the Seng Company an exclusive license to manufacture and sell under the pool patents. Under the terms of the contract, 33% of the total royalties was allotted to the Pullman Couch Company. Two companies and two independent inventors signed the licensing agreement: the Davoplane Bed Company (7 patents), the Pullman Couch Company (13 patents) and two inventors. In return, the Seng Company paid a fixed percentage to the pool, and then, pool members split the royalty according to a contractual formula. A few years later, in 1924, in the radio industry, an organization first-named the Associated Radio Manufacturers (renamed Radio Corporation of America) pooled the resources and technology of American Marconi, Westinghouse, General Electric, Telegraph (AT&T) and American Telephone. This pool would lead to the establishment of standardization of radio parts, airway’s frequency locations and television transmission standards.

The development of patent pools in the high-tech industry is emblematic in recent years. In 1997, several patentees of key technology created a patent pool in the video industry¹¹ and Philips Electronics N.V. to pool some essential patents related to the MPEG_2 compression technology standard.

¹⁰ Grover & Baker, I.M. Singer & Co., Wheeler, and Wilson & Co.

¹¹ The parties were the Trustees of Columbia University, Lucent Technologies Inc., General Instrument Corp., Fujitsu Limited, Matsushita Electric Industrial Co., Ltd., Mitsubishi Electric

Similarly, in 1998, Sony, Philips, and Pioneer formed a patent pool for technologies necessary to comply with some DVD-Video and DVD-ROM standard specifications. Finally, in 1999, Toshiba Corporation, Mitsubishi Electric Corporation, Hitachi, Ltd., Time Warner Inc., Matsushita Electric Industrial Co., Ltd., and Victor Company of Japan, Ltd. set up a patent pool to administer technologies manufactured in compliance with the DVD-ROM and DVD-Video formats.

Finally, sometimes cross-licensing may result out of a strategy initiated after a litigation settlement has taken place, so it becomes a preventive strategy for future claims. A case in point is the dispute between Nokia and Apple, analyzed above, whereby Apple agreed to pay Nokia an undisclosed amount for smartphone technology allegedly stolen. The agreement provided a one-off sum payable to Nokia followed by ongoing royalty payments. Apple noted this payment was the result of a cross-licensing agreement because it also considered its patent claims against Nokia and contended it did not include “the majority of the innovations that make the iPhone unique.”

Notwithstanding the clear advantages that patent pools bring about to firms that find it very expensive to commercialize their closely related patents separately, they do not eliminate financial risk in full, only temper it. Nevertheless, the pool does help to assure a common interest will emerge should one member be accused of infringement by a third party.

3.4 Patent Exchange Online Platforms

A clear problem arising out of the difficulty to measure IP value objectively became apparent in the failed attempts to develop broader IP market trade with the help of digital technologies.

Given the steady rise in the number of patent registries in the U.S. (and elsewhere) following the new perception of patents as a “core asset class” (Palfrey 2012), the conditions were seemingly ripe – or so it was thought – to institutionalize a comprehensive patent market exchange mechanism. Throughout the 2000s several companies built online platforms in an attempt to create more efficient ways to bring buyers and sellers of patents closer. These digital infrastructures were intended to facilitate transactions without taking ownership of the patents involved. Two types of platforms emerged: online marketplaces and live auctions.

Online marketplaces attempted to replicate the Alibaba or eBay model, by enabling better information about patent owners and technology sellers and potential buyers and licensees. Web sites such as Yet2 or Tynax list information of the patents that are for trading together with special conditions for the sale or licensing. Special tools give buyers an easy search of the valid patents both by patent classes and by keywords, as well as descriptions of specific intellectual assets of particular interest.

These platforms allow applicants to search without disclosing the buyers’ identity, coupled with curation carried out by the platform over the parties offering bids;

screening is critical to ensure the presence of active investors and to filter weak patents from accessing the platform. Confidentiality ensures that the parties will withhold their strategic competitive research advantage vis-à-vis other potential offering parties present in the platform. Moreover, trading partners hold no transactions online; rather, the platform connects buyers and sellers once the potential traders commit in a posting. Intermediaries appointed by the platforms then perform person-to-person deals. The platform charges commission fees for these services. In a nutshell, these platforms act as brokers for highly specialized patent deals.

However, high hopes were placed upon the capacity of technology to support patent trading through auction mechanisms administered by online platforms. Hagiu and Yoffie (2013) explain this mechanism:

These auctions functioned like other live auctions, for example, like art at Sotheby's and Christie's – with an auctioneer taking bids for each lot, which could be a single patent, copyright, trademark, or domain name right, or a bundle of such assets. The lots were sold to the highest bidder on condition that the highest bid exceeded the seller's reserve price.

Possibly IPXI was the most visible online auction mechanism for patent licenses, so it deserves closer inspection to understand the hurdles imposed by the patent system in allowing the emergence of efficient trading mechanisms.

This platform was launched in 2008 amidst high expectations that it would capitalize the richness of intellectual property scattered throughout the vast U.S. innovation ecosystem. IPXI expected to build an open and extended mechanism to help patent owners (particularly favoring SMEs) monetize their intellectual assets more efficiently and give them access to risk management tools to hedge their exposure. Its founders expected to develop a transparent marketplace for patent license exchanges that would lure investors to speculate on emerging technologies as opposed to investing in the equity of the companies that operate those technologies, with the ability to trade those rights as the value increased. In this way, IPXI expected to reduce investors' risk exposure.

IPXI's online auction trading platform could fulfill its harmonizing role as long as different trading parties could easily compare the information traded. To harmonize the information arising from specific technologies, IPXI created a new standard, the Unit Licence Right (ULR). This commodity or exchange-traded product offered purchasers a title to a license on a non-discriminatory basis, at a market-based price and could be sold on a standardized technology-unit basis that IPXI determined according to the underlying technology or IP of the ULR contract. Presumably, each ULR Contract offered its buyer a sound right to manufacture and sell a particular number of products or services employing the patented technology without the threat of an infringement suit. Thus, for example, a patent owner, i.e. licensor, could list on the IPXI a ULR contract that covered a patented nanotechnology for rapid detection of pathogens in organic liquids. The buyer of a ULR Contract could then manufacture diagnostic devices, use and sell the technology, or, a pre-defined number of devices with the patented nanotechnology. So, if the purchaser or licensee wanted to make and sell 500 devices and each ULR contract represented a single use of the IP, then the purchaser had to buy 500 ULR contracts. Each ULR

contract expired when one device was manufactured. A secondary market for unused ULR contracts was also expected to emerge.

IPXI divided available patent licenses into a defined number of ULRs; by slicing the implicit IP entitlements, potential investors could buy just the optimal number of ULRs they wanted to invest in; thus, the mechanism allowed them to hedge their risk more efficiently, as the system would minimize financial risk associated with idle ULR contracts. The IPXI published market-based pricing of the vetted technologies and presented potential investors with a URL standard-terms contract similar to that of public equity offerings of corporations. Moreover, buyers could sell ULRs if the licensee found it had too many; or it could buy more if necessary. IPXI offered a large market of ULRs to invest in, and that presented (at least in theory) a more efficient way of investing in emerging technologies than entering into murky, tortuous, and expensive bilateral deals.¹² Commentators at the time highlighted this point:

One of the biggest advantages of the ULR Contract for both IP licensees and licensors is the removal of the lawyer-intensive traditional bilateral licensing process (...) One of the IPXI's initial objectives was to build an exchange that operates under two core principles that do not exist in the traditional bilateral product/technology licensing scenario: transparency and efficiency. Regarding transparency, the IPXI starts the ULR Contract process by performing in-house a legal analysis designed to give the marketplace confidence in the quality of all patent rights listed as available ULR Contracts.

IPXI's business model was simple: both ULR contract sponsors, i.e. licensors, and investors, i.e. licensees, had to pay an annual membership fee of \$5000 each, and sponsors had to pay a \$100,000 listing fee per patents or patent portfolio. Also, IPXI's processing commission fee was set at 20% of the ULR Contract price.

Moreover, IPXI's founders put great effort into signing up members to the platform, to trigger initial use of the platform that could be followed by newcomers. Its membership included 70 leading innovative companies among the world's top owners of IP assets including several U.S. universities like Rutgers, Northwestern and the University of Utah. At some point, these institutions committed to issuing ULR Contracts with a market value of at least \$50 million (Boger and Ziegler 2012). Other sponsors included names such as Ford Global Technologies, MetaPower Inc., the University of Notre Dame, the Regents of the University of California, JP Morgan Chase and electronics giants Sony USA and Philips. Even the U.S. Department of Energy and various national laboratories committed to the IPXI as sponsors.

In short, all the odds seemed for IPXI's business model. The platform the auction mechanism provided was transparency to the licensing and created an open database of royalty rate data. Also, it standardized contracts used to do the patent licensing, thereby reducing transaction negotiation costs, and allowed parties to obtain licenses at comparable rates. Finally, it established a royalty database that could have allowed

¹² Critics had complained that IP licensing was an inefficient and costly way to do business. Most of these complaints centered on the fact that negotiations raised many transaction costs that outweighed the value of the license. These included value determination of the IP and the market potential of the technology; identifying possible licensees; once the licensee candidate was chosen, it was necessary to do a due diligence; and only then, negotiations over the license agreement conditions. In the most optimistic scenario, this process usually took several months to complete (Boger and Ziegler 2012).

third parties to increase their predictability about future potential licensing transactions or set damages in case of litigation. Moreover, it also provided access to technology users of all types: small companies, research organizations, and universities would have similar access to ULR contracts, as Fortune 100 companies. Finally, IPXI provided a rules-based approach to directed enforcement. What went wrong?

IPXI's model would prove more fragile than it looked on the surface. In his farewell memo, CEO Pannekoek told staff and members that the exchange was forced to close because the "time was just not right, and the obstacles – including potential licensees who challenged us to litigate instead of work with them – simply too insurmountable."

Given the lack of particular IP value, licensee candidates prefer to chase an expected "litigation price" that potentially yields them higher returns, as opposed to the "auction" price they are offered through standardized terms, such as the ULRs. IPXI was held up by the potential patent licensees' unwillingness to obtain required licenses unless threatened with litigation.¹³ It seemed obvious that licensees found it more lucrative to free ride on the technical innovation of the patentees, members of the IPXI's platform, than to pay royalties, no matter the curation services provided by the platform.¹⁴ Rent seeking presented them with a lower "litigation price," that yielded higher returns for the use of technology, than the amount of royalties calculated by IPXI's technicians. Even if ULR valuation included a significant price discount for litigation contingencies, and that legal due diligence was duly executed by supporting law firms, licensees were driven by the price potentially attainable in out-of-court settlements. Reality sank in, as everyone realized that given the murkiness of the IP system, betting on the judicial enforcement would be more lucrative than the hardships of negotiating IP prices.¹⁵

¹³ Pannekoek explained: We were astonished by some of the initial responses to our marketing efforts. We decided to go directly to the business operations of our potential licensees and approached CEOs and others in the executive suite to make a business case. When we sat down with those who got back to us, in most cases, the initial discussions went very well – people understood what we were saying about it being an efficient, transparent process which allowed the licensee to get fair terms based on a market price. But then, the process would stall. At some point, the door was shut and eventually the response was "we don't need to do anything until you file in court."

¹⁴ IPXI put together a package for a collection of nearly 200 IEEE standard 802.11n compliant patents. But as IPXI's CEO explained, "we provided in-depth due diligence on essentiality history, determination, claims charts and more, plus a 75 percent discount on payments relating to alleged past infringements, as well as the ability to be on a level playing field with market-based pricing," he explained. "But still they were not willing to transact or negotiate."

¹⁵ Contreras and Reichman (2015) argues that IPXI's demise was due to a complexity of factors. First, the patents whose licenses would align under ULRs were not essential to manufacturers, so IPXI would not serve as a one-stop solution for alleviating the burden of negotiating bilateral license agreements about the standard. Second, IPXI's sponsors were not necessarily perceived as key players in the licensees' market, nor did they have a tradition of license enforcers; hence, these manufacturers may not have seen these sponsors as a threat that could be forestalled through preventive licensing. Finally, and one could say, paradoxically, ULR standardization of contractual terms between licensors and licensees, which was intended to enhance market trading and pricing, was inflexible, hence unfit to accommodate specific transaction needs. Thus, ULRs were a less convenient option, compared to cross-licensing and patent pools which enable the licensee to use its patents as a currency in obtaining a desired license from the patent holder, thus offsetting part of their license cost.

IPXI's management failed to note that litigation is also another form of IP strategy that emerges out of the convoluted nature of the patent system. IP patent litigation is an inherent element of the IP system as much as a patent award because it is tough to value IP, so parties have incentives to leverage their market power if they have one. Far from the idealized Platonic assumption that people would eagerly submit their IP for an exchange forum like IPXI, he should have better started by questioning whether it would be more attractive for licensors to exploit their technology voluntarily, or to avoid exposure simply by not patenting their technology at all.

3.5 Patent Trading as IP Management Strategy

Patent trading remains the most important IP management strategy today, although its shape and goals are becoming more versatile as the popularity of competitively designed IP increases.

Given the pace of technological change, many companies are turning to new paradigms and models of innovation such as open and collaborative innovation. These models challenge the protection mindset and could give rise to new thinking and practices on IP ownership, sharing and licensing. An example is the 10-year-long broad patent cross-licensing deal recently signed between Google and Samsung. Patents may continue to increase, but the manner in which they are utilized and deployed are likely to evolve to accommodate the dynamism of the marketplace.

First, IP management today devises patent filing as a strategy to sense the market. A program of monitoring and reviewing filings that have the potential to influence one's space empowers companies to anticipate emerging threats and identify trends. This can support effective blocking strategies, such as defense patenting, whereby companies pursue IP in the unoccupied ground or white spaces where substitutes may emerge or where the competition may be going. Using analysis of patent filing activity can reveal where tomorrow's most important innovations will be gauged. This could help IP management strategists to anticipate where their competitors may be headed. For example, in terms of geography, since 2010 there have been five countries with the highest volume of filings: China, the U.S., Japan, Germany, and the Republic of Korea. Also, this data shows the relative specialization of countries in certain technologies: Japan in optics, the U.S. and China in digital communications.

Second, filing patents preemptively could be used to drive M&A activities, to block out current or potential competition, to create new business structures and partnerships or to establish a culture of innovation. For example, research on patent filing may clarify what technologies are "in" and what technologies are "out", thereby allowing businesses to plan their innovation strategies. For example, the recent increase of solar energy filings among patent applications in energy-related technologies in the U.S. shows that this technology is likely to be the prime technology dominating this market, above fuel cells, wind energy and geothermal.

Third, the convergence of technology acceleration and patent reform has created an urgent need for business leaders to reassess their IP strategies, to become much more assertive than in the past. By assertiveness we do not mean to be always on the offensive; defensive strategies are also part of the portfolio of IP management strategies. The point is, rather, that corporate leaders must acknowledge the nature of the battleground waged for the control of competitive capabilities. More and more, corporations fight in this war using a mix of technology and intellectual property. Players that do not understand this reality not only run the risk of becoming technologically obsolete from competitors they did not anticipate but also face the prospect of not being able to meet their challenge because patents have completely blocked them out of the game.

Of course, like any other market, IP trading needs information to function. However, it is evident that IP markets suffer from various forms of information asymmetry that undermine their transparency and place SMIEs in a delicate position, as they can hardly obtain information because of their comparative weaker financial resources to obtain such missing information about clients, interested investors, complementary technologies, cost-effective jurisdictions, where to avoid being sued by a stronger party, and so on. If the potential licensor and licensee knew how much the technology involved in the deal was worth, they could quickly reach an efficient agreement that would be mutually beneficial in the absence of transaction costs (Coase 1960). This is far from the case. IP market murkiness is geared against SMIEs because they lack resources to obtain such information. Their weak position enhances the impact of transaction costs arising out of the lack of transparency featuring IP markets.

Institutions emerge to fill this gap by allowing IP trading parties to communicate information effectively. This conclusion sounds simpler than what it is. As IPXI's failure shows, the inherent murkiness of the IP patenting system tends to prevail upon the efforts across the board aimed at creating stable IP market prices. Participants in the IP market preferred to litigate than coalescing into the information mechanisms provided for by the IPXI market. Clearly, this is a case where the Coasean transaction costs inherent in the underlying discretion within which patents are afforded it impossible to reach the optimal solution that an alternative institutional arrangement would have brought about.

In general, these platforms failed to allow trading to occur, and its promoters quickly migrated to consulting services or brokerage.¹⁶ Online marketplaces, as well as auction platforms, proved unviable due to the complexities and information asymmetries of the IP market. These asymmetries forced platform managers and hosts to become advisors and brokers supplying the market with information, as the working of these infrastructures alone failed to encourage transacting parties to disclose such information by themselves. Naturally, that undermined the business model implicit in the very concept of platforms, namely, to promote exponential growth by minimizing entry thereby easing curation mechanisms.

¹⁶These include platforms such as Yet2 (<http://www.yet2.com/>) and Tynax (<http://www.tynax.com/>).

In the IP markets, particularly patent trading would prove too complicated for the trading partners to assume this activity by themselves. Searching for the “right” technology, assessing the legal title of those claiming to be owners of that technology, and the hurdles implicit in facilitating the legal exchange¹⁷ could prove enormous.

Very often, IP players apply business tactics intended to leverage the system in their favor to tap the information gaps created by the IP system. Thus, it seems that the development of IP strategies in the U.S. does also include that of making a trade-off between the amounts of information that the innovator will be interested in disclosing in a patent application, versus staying quietly on the sidelines, trying not to be noticed. The excess of litigation indicates a very responsive system that highlights the gaps of information accruing from its failure to assign IP claims in a non-contested manner. Let us turn our attention to that problem in the next chapter.

In the case of Latin America, we associate the cognitive problem arising from the use of intellectual assets with a failure to understand the system, but in a different way; rather than availing from the legislation to exert rents through abusive litigation, in Latin America innovators seem to adopt the opposite direction.

¹⁷These difficulties include coordination costs between potential co-owners, coordinating costs with the sellers assessment of potential market success for the technology and IP valuation.